



application performance, detect issues, and optimise resource usage. Observability practices include setting up monitoring dashboards, analysing metrics, and configuring alerts to ensure system health.

In 2025, observability is foundational for managing scalable and resilient applications. Developers should familiarise themselves with open source observability tools, practice analysing performance data, and design alerting systems to proactively manage potential failures. These practices improve application quality and help maintain seamless user experiences.

Security: Protecting applications against evolving threats

With increasing cyber threats, robust security measures are indispensable for protecting modern software systems. Open source tools like OWASP ZAP and Clair (for container security) offer vulnerability scanning, while platforms like Wazuh provide comprehensive monitoring, intrusion detection, and log analysis capabilities. These tools empower developers to detect, mitigate, and respond to security incidents efficiently. Best practices such as automated security testing and regular assessments help mitigate risks. Incorporating security measures during development ensures that applications remain resilient and trustworthy.

In 2025, prioritising security helps safeguard sensitive data and builds user confidence. Developers can get started by integrating automated scanning tools into their CI/CD pipelines, deploying Wazuh for proactive monitoring, and staying updated on emerging security threats. Regular vulnerability assessments and adopting industry-standard encryption methods further enhance application security.

In 2025, open source technologies like containerisation, microservices with serverless computing, generative AI, API-driven development, robust IAM systems, observability, and security are defining the future of software development. For college students majoring in computer science or related fields, these areas offer exciting opportunities to innovate, grow, and shape the digital world. Begin exploring these technologies today to gain hands-on experience, join vibrant developer communities, and position yourself as a key player in the evolving tech landscape. **END** 🐼

By: Deepen Dhulla

The author has over 25 years of experience in Linux and open source technologies. He now leverages his expertise to guide organisations in designing resilient IT infrastructures and innovative solutions.

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